

Oblicz $P(A|B)$, jeśli wiadomo, że $P(A \cup B) = \frac{3}{4}$, $P(B) = \frac{2}{3}$, $P(A) = \frac{1}{4}$

① DANE:

$$\begin{cases} P(A) = \frac{1}{4} \\ P(B) = \frac{2}{3} \\ P(A \cup B) = \frac{3}{4} \end{cases}$$

wzór na prawdopodobieństwo warunkowe

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

② liczymy $P(A \cap B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\begin{aligned} \Rightarrow P(A \cap B) &= P(A) + P(B) - P(A \cup B) = \\ &= \frac{1}{4} + \frac{2}{3} - \frac{3}{4} = \frac{3+8-9}{12} = \frac{1}{6} \end{aligned}$$

③ podstawiamy wyliczone $P(A \cap B)$ do wzoru

$$P(A|B) = \frac{\frac{1}{6}}{\frac{2}{3}} = \frac{1}{6} \cdot \frac{3}{2} = \frac{1}{4}$$

odp: $\frac{1}{4}$